We remark that if a "point at infinity" is adjoined in each coordinate (one-point compactification of VF) then there is no need to require that A be bounded.

revision of the next few lemmas

Unapplied R040/R165 proposal for author review. Put VF-generation in this lemma as a local hypothesis. The original algebraic-closure model argument needs it: a ground base with an extra residue-transcendental RV parameter need not have VF representatives of that parameter. Spell out $A \subseteq \mathrm{VF}^n$ and definability, as needed by later applications to graphs. Require finite nonempty values and the singleton limit already delivered by the construction. Use projective charts for infinite coordinates and extend the continuous germ by a constant finite nonempty set on the clopen complement, avoiding the scalar-rescaling issue.
The two later geometric applications (closed bounded images and the closed-graph criterion) need not acquire a standing VF-generation restriction: their proofs may temporarily enlarge the parameter base before choosing curves, since their conclusions request no choices definable over the original base. Their separate proof revisions remain to be reviewed; no global parameter assumption is changed by this draft.

Lemma 2.46 (Curve selection)**.** *Suppose that $\mathbb{S}$ is VF-generated. Let $A \subseteq \mathrm{VF}^n$ be definable and let b be a point in its closure in the coordinatewise projective line $(\mathbb{P}^1(\mathrm{VF}))^n$, with the valuation topology. Then there is a b-definable continuous finitary function $g : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ whose values are nonempty and such that*

$$\lim_{\mathcal{M}\setminus\{0\} \longrightarrow 0} g = \{b\}.$$

Referee R165 [Clarification; new] – Definability and nonempty values in curve selection. *Original source line 1081.* State that A is definable. In the proof require $g(c) \neq \varnothing$ for every c (or at least on a punctured neighborhood of 0): "finitary" in the earlier definition allows empty values, and $g(c) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$ alone is also satisfied by the identically empty correspondence. The algebraic-closure construction, once its base hypothesis is corrected below, supplies finite nonempty values and hence the stronger conclusion $\lim g = \{b\}$.

Proof. Use a coordinatewise projective chart around b to reduce to the case $b \in \mathrm{VF}^n$: at an infinite coordinate use inversion and restrict A to the chart. These changes are definable over b and preserve convergence and continuity locally.

Naming the finite tuple b and a nonzero $c \in \mathcal{M}$ preserves VF-generation and the standing nontriviality of the ground valuation. Hence $\mathrm{acl}(\mathbb{S}, b, c)$ is a model of $\mathrm{ACVF}(0,0)$. Since $b \in \overline{A}$, the definable set $A \cap \mathfrak{o}(b, \mathrm{val}(c))$ is nonempty, so elementarity gives a point of that intersection algebraic over $(\mathbb{S}, b, c)$. Compactness supplies a b-definable finitary function g_0 on $\mathcal{M} \setminus \{0\}$ such that

$$\varnothing \neq g_0(c) \subseteq A \cap \mathfrak{o}(b, \mathrm{val}(c))$$

for every c. Thus g_0 converges to $\{b\}$, and nonempty values make this limit minimal.

By Lemma 2.44, with its one-dimensional source, g_0 is continuous outside a finite set. Choose a b-definable clopen disc $D \subseteq \mathcal{M}$ around 0 such that g_0 is continuous on $D \setminus \{0\}$. Also $\mathrm{acl}(\mathbb{S}, b)$ is a model, so A contains a finite nonempty b-definable set E. Define

$$g(c) = \begin{cases} g_0(c), & c \in D \setminus \{0\}, \\ E, & c \in \mathcal{M} \setminus D. \end{cases}$$

The two pieces are relatively clopen in $\mathcal{M} \setminus \{0\}$, so g is continuous, finite-valued and nonempty everywhere. Its limit at 0 is still $\{b\}$. Finally transport the construction back through the projective

chart. This avoids assuming that a definable value-group radius is the valuation of a definable scalar. □

Referee R040 [Error] – Curve selection requires a restriction on the base.
Original source line 1088. The assertion is false for the arbitrary $\mathbb{S}$ allowed so far. Let K be a nontrivially valued algebraically closed base field, add an RV parameter t with $\mathrm{rv}(t) = 0$ transcendental over its residue field, and put $A = \{(x, y) : y \neq 0, \mathrm{rv}(x/y) = t\}$. Then $(0, 0) \in \overline{A}$. A nonempty finite $\mathbb{S}$-definable choice in A at an input $c \in K \cap (\mathcal{M} \setminus \{0\})$ would have coordinates algebraic over K, t, c, hence in K by Lemma 2.13; their ratio cannot have leading term t. Even though finitary values may be empty, $b \in \lim g$ forces their support to accumulate at 0; by C-minimality that support contains a punctured neighborhood, whose defining radius lies in $\Gamma(K)$ since adjoining the residue parameter t does not enlarge definable values. It therefore contains such an input $c \in K$. Thus the claimed curve does not exist over that base. Impose VF-generation before this lemma (as already suggested in the human report), or allow explicitly specified extra VF representatives. Under VF-generation, the added nonzero $c \in \mathcal{M}$ makes the algebraic closure a model, and the displayed construction yields nonempty finite values converging to b itself. For the last rescaling step, first shrink to a disc whose radius is the valuation of a definable scalar; the original definable radius need not be realized by such a scalar.

This is an analogue of the highly useful o-minimal curve selection.

Remark 2.47. Although the Γ-sort is essentially an o-minimal group, curve selection fails due to the presence of the top element ∞. For example, the set $\Gamma \times (0, \infty) \subseteq \Gamma_\infty^2$ contains curves approaching the boundary point $(0, 0)$ or (∞, ∞) but no curves approaching the boundary point $(\infty, 0)$, because all curves are lines.

Referee R166 [Typo; new] – Eventually affine curves in the value group.
Original source line 1094. Replace "all curves are lines" by "every definable one-parameter curve is piecewise rational-affine". Near a one-sided endpoint, one affine piece remains; a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞. This is the precise reason the stated boundary point cannot be approached.

Lemma 2.48. *Let A be a closed and bounded set in* VF*, and $f : A \longrightarrow \mathrm{VF}^m$ a definable continuous function. Then $f(A)$ is also closed and bounded.*

Proof. Suppose for contradiction that $f(A)$ is not closed and bounded. Applying curve selection to any point in the frontier of $f(A)$, we obtain a curve $C \subseteq f(A)$ such that $\lim_{\mathcal{M}\setminus 0\to 0} C$ is not contained in $f(A)$. Since $f^{-1}(d)$ contains d-algebraic points for every $d \in f(A)$, we can pull back C along f, that is, we can find a curve $C' \subseteq A$ with $f(C') = C$. By Lemma 2.45, $\lim_{\mathcal{M}\setminus 0\to 0} C' \subseteq A$. Since f is continuous, this would mean $\lim_{\mathcal{M}\setminus 0\to 0} C \subseteq f(A)$, contradiction. □

Referee R041 [Gap] – Selecting and lifting the limiting curve.
Original source line 1104. If the image is unbounded but closed in affine space, its affine frontier need not contain a point outside it. Take the frontier in the coordinatewise projective line and allow an infinite coordinate. Also the existence of algebraic points in every fiber needs justification under the chosen parameter assumptions; for the curve pullback it is enough to include the curve parameter c in the algebraic closure and apply the model argument there. Preserve a nonempty finite-valued parametrization throughout and apply the limit lemma in its multidimensional form.

Corollary 2.49. *If the function f in Lemma 2.48 is injective then it is a homeomorphism from A onto $f(A)$.*